

Question 5

New Jersey

The new animated feature film is now playing 3 times a day in DHA Cinema. One day, there were 20 adults, 30 children and 10 senior citizens and theater made \$600. At the next showing there were 20 adults, 48 children and 20 senior citizens with the theater making it \$800 in ticket sale. At the last showing theater made \$400 with 10 adults, 30 children and 5 senior citizens. How much are the tickets at the movie theater?

	I	II	III
A \$x	20	20	10
C \$y	30	48	30
S \$z	10	20	5
	600	800	400

$$\begin{array}{rcl}
 20x + 30y + 10z & = & 600 \quad \div 10 \\
 20x + 48y + 20z & = & 800 \quad \div 4 \\
 10x + 30y + 5z & = & 400 \quad \div 5
 \end{array}$$

$$2x + 3y + z = 60$$

$$5x + 12y + 5z = 200$$

$$2x + 6y + z = 80$$

$$2(3) - 5 = 1$$

$$3R_1 - R_2 \left(\begin{array}{ccc|c} 2 & 3 & 1 & 60 \\ 5 & 12 & 5 & 200 \\ 2 & 6 & 1 & 80 \end{array} \right)$$

$$3R_1 = \begin{array}{rrrr} 6 & 9 & 3 & 180 \\ -5 & 12 & 5 & 200 \\ \hline 1 & & & \end{array}$$

$$\left(\begin{array}{ccc|c} 1 & -3 & -2 & -20 \\ 5 & 12 & 5 & 200 \end{array} \right)$$

$$\left(\begin{array}{ccc|c} + & - & & \\ \hline 5 & 12 & 5 & 200 \\ 2 & 6 & 1 & 80 \end{array} \right)$$

$$\begin{array}{l} R_2 - 5R_1 \\ R_3 - 2R_1 \end{array} \left(\begin{array}{ccc|c} 1 & -3 & -2 & -20 \\ 0 & 27 & 15 & 300 \\ 0 & 12 & 5 & 120 \end{array} \right)$$

$$R_2: 5 \mid 12 \quad 5 \quad 200$$

$$5R_1: -5 \quad -15 \quad -10 \quad -100$$

$$\begin{array}{r} 0 \quad 27 \quad 15 \quad 300 \\ \hline R_2 \quad 2 \quad 6 \quad 1 \quad 80 \\ 2R_1 \quad 2 \quad -6 \quad -4 \quad -40 \\ \hline \end{array}$$

$$\frac{1}{27} R_2 \left(\begin{array}{ccc|c} 1 & -3 & -2 & -20 \\ 0 & 1 & \frac{15}{27} & \frac{300}{27} \\ 0 & 12 & 5 & 120 \end{array} \right)$$

$$\begin{array}{l} a_1 x + b_1 y = C_1 \\ a_2 x + b_2 y = C_2 \end{array}$$

Homogeneous Equation

$$C_1 = C_2 = 0$$

$$\left. \begin{array}{l} a_1 x + b_1 y = 0 \\ a_2 x + b_2 y = 0 \end{array} \right\} \begin{array}{l} (0,0) \\ \text{Trivial} \\ \text{Solution} \end{array}$$

Consistent equations

(soln)

unique / infinite

$$\frac{a_1}{a_2} \neq \frac{b_1}{b_2} \quad \text{unique soln}$$

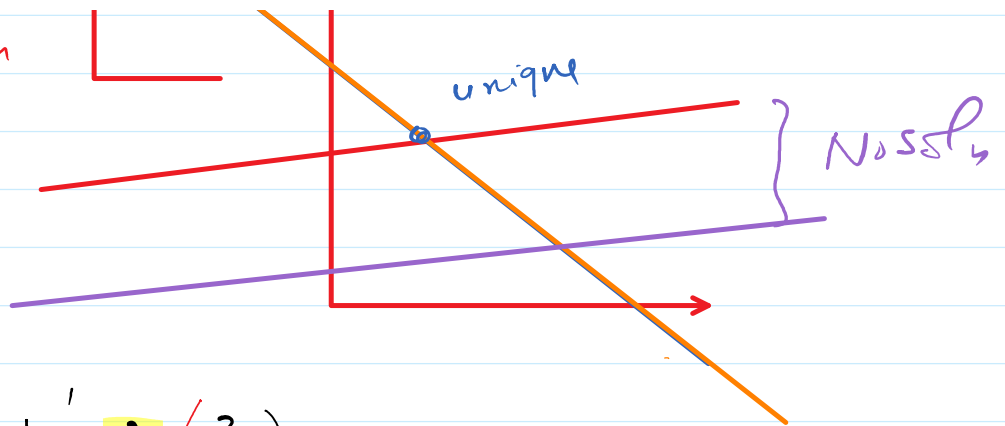
$$\frac{a_1}{a_2} = \frac{b_1}{b_2} \neq \frac{C_1}{C_2}$$

$$\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{C_1}{C_2} \quad \text{infinite soln}$$

→ infinite

unique

infinite solⁿ



$$A = \begin{pmatrix} -1 & 2 & 3 \\ 4 & 2 & 1 \\ -3 & 2 & -1 \end{pmatrix}$$

$$|A| = a_{ij} c_{ij} \quad \text{let } i=3$$

$$= a_{3j} c_{3j}$$

$$= a_{31} c_{31} + a_{32} c_{32} + a_{33} c_{33}$$

$$= -3 \times -4 + 2(13) + (-1)(-10) = 12 + 26 + 10 = 48$$

$$C_{ij} = (-1)^{i+j} M_{ij}$$

$$C_{31} = (-1)^{3+1} M_{31} = +1 \left| \begin{array}{cc|c} 2 & 3 \\ 2 & 1 \end{array} \right| = 2 - 6 = -4$$

$$C_{32} = (-1)^{3+2} M_{32} = -1 \left| \begin{array}{cc|c} -1 & 3 \\ 4 & 1 \end{array} \right| = -1(-1 - 12) = 13$$

$$C_{33} = (-1)^{3+3} M_{33} = +1 \left| \begin{array}{cc|c} -1 & 2 \\ 4 & 2 \end{array} \right| = (-2 - 8) = -10$$

Cryptography

Ciphering & deciphering

$$KP = C$$

Modul 26

Reciprocal

+

$$26 =$$

$$28 \equiv 2 \pmod{26}$$

$$1070 \equiv 4 \pmod{26}$$

$$26 \overline{) 1070} = 41 \dots$$

$$41 \times 26 = 1066$$

$$10 + 520 = 530$$

(-ve)

$$-4 \equiv 22 \pmod{26}$$

$$-26 \equiv 26 \pmod{26}$$

$$+26$$

$$0 + 26$$

$$-40 \equiv 12 \pmod{26}$$

$$26 \overline{) 40}$$

$$20 \times 2$$

$$(- \div 26)$$

$$-40$$

$$\underline{+26}$$

$$-14$$

$$\underline{+26}$$

$$12$$

$$-40 + 52$$

$$26/1070 = 41 \dots$$

$$41 \times 26 = 1066$$

$$-510 + 520 = 10$$

$$19 \dots$$

$$26 \overline{) 40}$$

$$20 \times 2$$

$$\begin{array}{r} -40 \\ 26 \\ \hline -14 \\ +26 \\ \hline 12 \end{array}$$

Reciprocals

$$\frac{1}{3} \equiv 9 \quad (\div 26)$$

3K $\rightarrow$ Remainder 1

$$3 \times 9 = 27 \quad (1)$$

$$\frac{1}{9} \equiv 3? \quad (26)$$

$$\frac{1}{5} \equiv (K) 21$$

$$5K \Rightarrow 26 \times 4 = 104$$

$$5 \times 21 = 105$$

∞ 1 3 5 7 9 11 15 17 19,

a^{-1} 1 9 21

$$\frac{1}{3}$$

$$K P = C$$

$$P = \bar{K}' C$$

$$\bar{K}' = \frac{1}{(K)} \text{adj } K$$

$$K: \begin{pmatrix} 2 & 5 \\ 3 & 3 \end{pmatrix} \begin{pmatrix} W & O & G \\ R & N & G \end{pmatrix} \text{ WRONG}$$

$$\frac{69}{54} \div \frac{1}{3}$$

$$\begin{pmatrix} 2 & 5 \\ 3 & 3 \end{pmatrix}$$

$$\begin{pmatrix} 23 & 15 & 7 \\ 18 & 14 & 7 \end{pmatrix}$$

$$= \begin{pmatrix} 136 & 100 & 49 \\ 123 & 87 & 42 \end{pmatrix} \div 26$$

$$= \begin{pmatrix} 6 & 22 & 23 \\ 19 & 9 & 16 \end{pmatrix}$$

$$|A| = \begin{vmatrix} 6 & -15 \\ -9 & 2 \end{vmatrix} = 12 - 135 = -123$$

$$\text{adj. } A = \begin{pmatrix} 2 & -5 \\ -3 & 6 \end{pmatrix}$$

$$= \begin{pmatrix} 19 & 9 & 16 \\ \text{FSVIP} \end{pmatrix}$$

$$A^{-1} = -\frac{1}{-123} \begin{pmatrix} 2 & -5 \\ -3 & 6 \end{pmatrix} = \frac{1}{123} \begin{pmatrix} 2 & -5 \\ -3 & 6 \end{pmatrix}$$

$$= \frac{1}{123} \begin{pmatrix} -3 & 5 \\ 3 & -2 \end{pmatrix}$$

$$A^{-1} = \begin{pmatrix} -9 & 15 \\ 9 & -6 \end{pmatrix}$$

$$= \begin{pmatrix} 17 & 15 \\ 9 & 20 \end{pmatrix}$$

$$\begin{pmatrix} 17 & 15 \\ 9 & 20 \end{pmatrix} \begin{pmatrix} 6 & 22 & 23 \\ 19 & 9 & 16 \end{pmatrix} = \begin{pmatrix} 387 & 509 & 631 \\ 4 & & \end{pmatrix}$$